

Anvisningar/Instructions

Individuella inlämningsuppgifter:

Du väljer ett projekt från följande lista, genomför det, och skriver en rapport som ska lämnats in före slutdatum. I din rapport, ska det ingå detaljer av genomförande, resultat med tabeller eller grafer som bevis (vilket kanske inte är tillämpligt i vissa projekt), och slutsats. Du får skriva på svenska eller på engelska. Det finns ingen mall för skrivning. En riktlinje är att man ska skriva ungefär 5-6 A4 sidor (font 12). Några kanske behöver skriva mer och det är ok.

Individual assignments:

You select one project from the following list, implement it, and write a report. In your report, it should include details of implementation, results with tables or graphs as evidence (which might not be applicable in some projects), and conclusion. You can write in Swedish or in English. There is no template for the writing. A guideline is to write about 5-6 A4 pages (font 12). Somebody might need more pages and it is ok.

Project 1 (Chapter 1, P-5 in the textbook, pp. 53): A Toy Cryptosystem

Implement a toy symmetric cryptosystem based on the following method.

- a. Keys are 16-bit values.
- b. Messages are strings with an even number of characters (and each character is a 8-bit value using ASCII coding). One can always add a blank at the end of an odd-length string.
- c. The encryption of a message M of length n (in bytes) is given by

$$E_K(M) = M \oplus (K \| K \| \dots),$$

where the key K is repeated $n/2$ times.

- d. The decryption algorithm for a ciphertext C is the same as the encryption algorithm:

$$D_K(C) = C \oplus (K \| K \| \dots).$$

Implement a brute-force decryption attack for this cryptosystem and test it on randomly generated English text messages. Automate the process of detecting whether a decrypted message is English text.

Project 2 (Chapter 2, P-3 in the textbook, pp. 111): A Demo ATM

Design and implement a program simulating the main security functions of an ATM machine. In particular, your system should authenticate users based on a PIN and should transmit data to the bank in encrypted form. You should reduce the sensitive information stored by the ATM machine in between transactions to a minimum.

Project 3 (Chapter 2, P-4 in the textbook, pp. 111): Privacy Concerns on Using RFIDs

Write a paper that discusses the different kinds of RFIDs, including both self-powered and not. Address privacy concerns raised by wide-spread RFID use, such as in e-passports. Use research articles available on the Internet as source material.

For this particular task, you need to cite at least 6 academic articles (and thus not from Wikipedia or newspapers) regarding RFID and its security concerns. You also need to propose a solution to the RFID security issues. It is recommended to use the university library resources (where you have access to the ACM database, IEEE database, and some other databases).

Project 4 (Chapter 3, P-7 in the textbook, pp. 172): Buffer Overflow Attacks

Write a paper describing how buffer overflows are used as vectors for many computer attacks. Discuss how they enable different kinds of attacks and describe how different software engineering practices and languages might encourage or discourage buffer-overflow vulnerabilities.

For this particular task, you need to cite at least 6 academic articles (and thus not from Wikipedia or newspapers) regarding buffer overflow and its security concerns. You also need to suggest a good software engineering practice to address the buffer overflow issue. It is recommended to use the university library resources (where you have access to the ACM database, IEEE database, and some other databases).

Project 5 (Chapter 4, P-3 in the textbook, pp. 220): Worm Propagation Simulation

Write a simulator that can track how a computer worm propagates in a network of 1 million computers, such that n of these computers are vulnerable to this particular worm. In each step of the simulation, each infected computer randomly picks d other computers and tries to infect them. If a computer is attacked, then it is infected only if it is vulnerable. And if an infected computer is attacked, it will be reinfected according to a random reinfection probability, p . Run a number of experimental simulations for various values of the parameters n , d , and p , including the cases $p=0$, $p=1/2$ and $p=1$, keeping track of how many infections and reinfections occur on each vulnerable computer, as well as the total numbers of each category of vulnerable computer. Find the parameter values that cause the worm propagation to die out after a few rounds without infecting all the vulnerable computers and also find parameter values that cause the worm to overload the vulnerable computers.

In the above question, all worm copies will try to infect other computers at every round. There is an important parameter that was not mentioned explicitly, and that is the recovery rate r . A worm infection can be removed with certain probability (with recovery probability 1, it will mean that an infected computer will only be able to infect other vulnerable computers during one round first after infection and go back to the vulnerable state thereafter). You define this probability by yourself. This probability can be worm dependent (for each worm copy) or computer dependent. In this question, we do not consider patching or other type of remedies which will permanently turn a vulnerable computer invulnerable.

‘Overloading’ means all (or a high proportion of) vulnerable computers are going to be infected and will continue to be infected (despite recovery possibility) and the total number of infected computers do not drop.

For this particular task, you need to show the worm propagation graph (i.e. the number of infected computers at different time/round) with different parameters.

Project 6 (Chapter 4, P-4 in the textbook, pp. 220): A Business Model for Adware

Write a paper that discusses the business model for adware. Use articles from the university library and the Internet as source material for this paper. Include in your paper the risks, benefits, and costs for advertisers, adware designers, and the people who run adware servers.

For this particular task, you need to cite at least 6 academic articles (and thus not from Wikipedia or newspapers) regarding adware and its business model. At the end of the paper, you also need to suggest a countermeasure to address the adware issue from a business or technical point of view. It is recommended to use the university library resources (where you have access to the ACM database, IEEE database, and some other databases).

Project 7 (Chapter 5, a modification of P-8 in the textbook, pp. 268): Sniffing Wi-Fi

This is an individual work.

Find a Wi-Fi access point and access it with at least two computers (or mobile phones), one of which has packet-sniffing software (e.g. Wireshark) installed. Have one computer access the Internet using various tools, such as browsers and email clients, and have another computer watch their packets. Write a report describing the session (with graphs), including the issues of privacy and security that it raises. Pay attention that Wi-Fi traffic monitoring is not supported by default on Windows for many tools.

For this particular task, you need to cite at least 6 academic articles (and thus not from Wikipedia or newspapers) regarding packet sniffing and Wi-Fi security. It is recommended to use the university library resources (where you have access to the ACM database, IEEE database, and some other databases).

Project 8: (Modified from Chapter 6, P-5 in the textbook, pp. 325): A Linux Firewall Configuration

Most Unix-based operating systems have a firewall application which is called *Iptables*. Iptables can be configured to filter IP packets and is quite flexible. Develop a list of firewall rules that need to be enforced on the machine, and configure *Iptables* to enforce these rules.

Step 1: Create a Linux (or similar) environment. If you do not use Linux regularly, you can easily install a virtual machine using Virtual Box or VMware.

Step 2: Install Iptables if it is not installed by default. It will require root privilege. You can do like:
sudo apt-get install iptables

Step 3: Learn to configure *Iptables* so that only web traffic are allowed through. Verify the effectiveness of your configuration. Pay attention to the default firewall policy (i.e. ACCEPT, REJECT or DROP). In order to allow a web visit to a URL address, you would also need to allow the passing through of DNS traffic.

Step 4: Learn to configure *Iptables* so that only outgoing web visits are allowed through. In addition, the firewall should be stateful in the sense that only incoming web traffic that are associated with an already established connection should be allowed through. Verify the correctness of your configuration.

Step 5: Permanently save your *Iptables* configuration and print out your configuration in your report. By default, the *Iptables* configuration are flushed after each reboot.

Project 9 (Chapter 7, P-1 in the textbook, pp. 384): Privacy and Legitimacy Concerns for Cookies

Write a paper that describes the privacy and legitimacy concerns of cookies. As a part of this paper, you should use a web browser that allows users to examine the cookies that are stored in that browser. Begin by deleting all your cookies and then visit a popular news, shopping, social networking, or other web sites, to determine which cookies are set by that site. Examine these cookies to see what information they hold and write about the implications.

For this particular task, you need to cite at least 6 academic articles (and thus not from Wikipedia or newspapers) regarding cookies and their security. You also need to show some screenshots from your own experiment described above. It is recommended to use the university library resources (where you have access to the ACM database, IEEE database, and some other databases).

Project 10 (Chapter 7, P-3 in the textbook, pp. 385): A Demo of XSS Attack

On your home network or on an authorized virtual machine network, define three (virtual) machines, Web Server, Victim, and Attacker, which could in fact all be on the same host computer. On Web Server, install a web server software package (e.g. Apache) and a given web application that is vulnerable to the XSS attack (and you need to find such an application or develop one by yourself). Have attacker visit the vulnerable Web Server and inject malicious contents to Web Server, such that when Victim visits the infected web page, Victim is tricked to do things that is against its will.

Read the book for more details of XSS attacks. In your report, write details of your experimental setup and demonstrate the result from a successful XSS attack.

Project 11 (Chapter 8, P-2 in the textbook, pp. 443): AES Implementation

Write a program that can perform AES encryption and decryption. You can use whatever languages and libraries for this implementation. A recommendation is to use Java and Java Cryptography Architecture.

For the demonstration purpose, please encrypt the message “*Introduction to Computer Security*” and then decrypt it using a key of your choice. In your report, it is necessary to provide your source code and your annotations to the code.

